NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-27/70B
11 July 1979
SUPERSEDING
MIL-T-27/70A
18 May 1977

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY TF5S21ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.

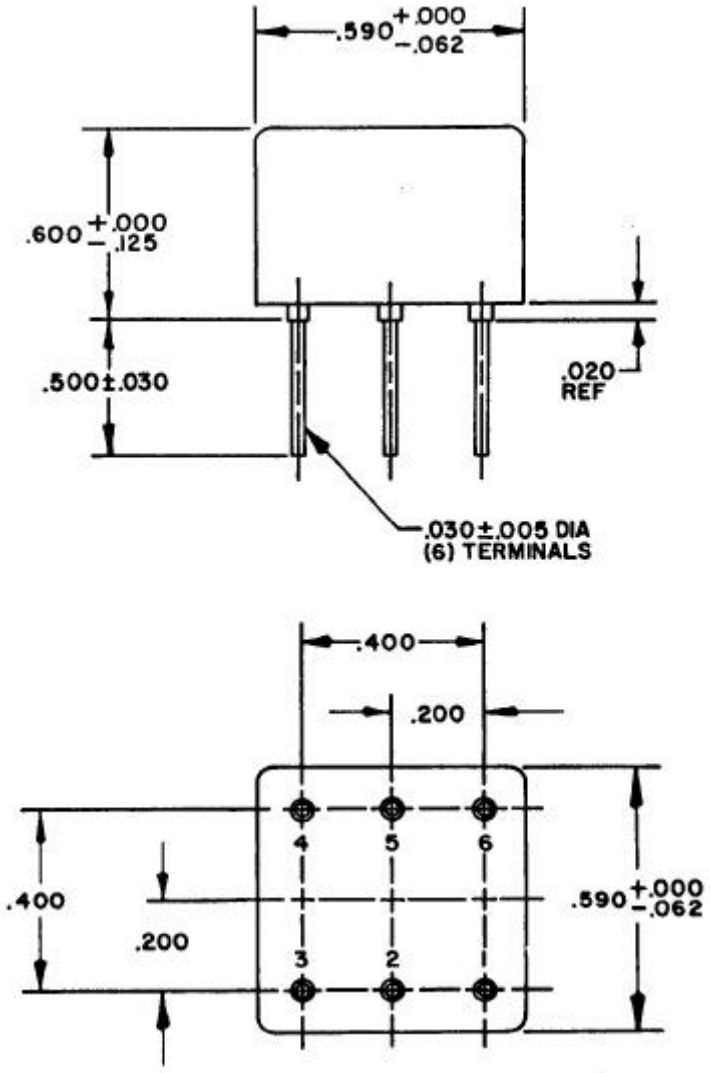
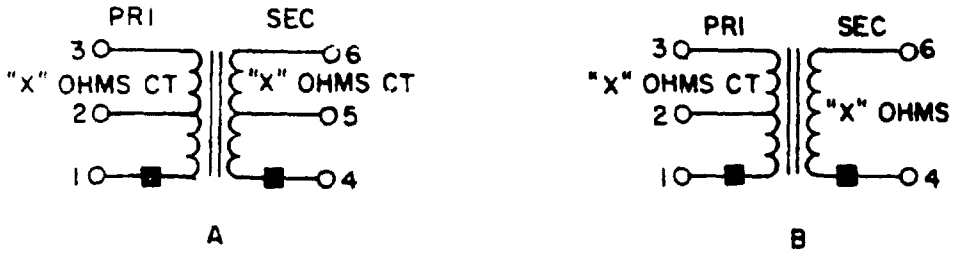


FIGURE 1. Dimensions and configuration.

FSC 5950

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INCHES	MM
005	13
020	.51
030	76
062	1 57
125	3 18
200	5 08
400	10 16
500	12 70
590	14 99
600	15 24



WORKING VOLTAGE. 175 V
ALTITUDE 70kFT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAMS AND MARKING

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3. Unless otherwise specified, tolerance is ± 005 (13 mm)
4. Marking shall be on sides and/or top of case
5. Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS: (When numbers in parentheses, i.e. (1-3) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.
Working voltage: 175 volts.
Frequency range: 200 hertz to 50 kilohertz (except 200 hertz to 15 kilohertz for M27/70-01)

TABLE I. Electrical ratings.1/

Dash No.	Circuit diagram (see figure 1)	Primary impedance ohms (1-3)	Secondary impedance ohms (4-6)	Primary DCR ohms ±25%	Primary power level at 200 Hz mW (max)	Secondary DCR ohms ±25%	Primary unbalanced DC current max mA
01	A	100 k CT	1,000 CT	5,300	60	55	0
02	A	25 k CT	1,000 CT	2,600	150	130	0
(B) 03	A	10 k CT	10 k CT	1,000	150	1,300	1.0
04	A	10 k CT	1,500 CT	1,000	150	200	1.0
05	A	10 k CT	600 CT	1,000	150	65	1.0
06	A	4,000 CT	10 k CT	350	150	1,300	1.5
07	A	1,000 CT	600 CT	90	150	65	3.0
(B) 08	B	1,000 CT	4	90	150	0.5	3.0
09	A	600 CT	25 k CT	50	150	3,200	4.0
10	A	600 CT	600 CT	50	150	65	4.0
11	B	600 CT	60	50	150	6.5	4.0
12	B	600 CT	4	50	150	0.5	4.0

1/ Qualification testing to M27/70-01 and all tests per table VII of MIL-T-27 for M27/70-05 and M27/70-08 shall be sufficient to grant qualification approval to M27/70-01 through M27/70-12.

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case Encapsulated.
Material: Plastic.
Terminals: Pin (printed circuit).
Material: Nickel iron alloy tin plated
Diameter 0.030 ±0.005 inch.
Length: 0.500 ±0.030 inch.
Weight 12.8 grams maximum.
Altitude. 70,000 feet maximum.
Operating temperature range: -55°C to +130°C.

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:
At sea level 500 volts rms.
At barometric pressure: 300 volts rms.

Electrical characteristics See table II.
Frequency response:
Reference frequency 1 kilohertz.
Polarity Additive, with terminals 3 and 4 connected.

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TABLE II Electrical characteristics.

Dash No	Frequency response at 200 Hz to 50 kHz ±1 dB			
	Z_S (1-3) ohms	Z_L (4-6) ohms	E_L volts	Primary DC current (1-3) mA
01 1/	100 k	1000	7.0	0
02	25 k	1000	7.0	0
03	10 k	10 k	22.0	0.5
04	10 k	1,500	22.0	0.5
05	10 k	600	22 0	0.5
06	4000	10 k	14.0	0.75
07	1,000	600	7.0	1.5
08	1,000	4	7.0	1.5
09	600	25 k	5.0	2.0
10	600	600	5.0	2.0
11	600	60	5.0	2.0
12	600	4	5.0	2.0

1/ Frequency response is 200 hertz to 15 kilohertz ±1 dB.

Marking location See figure 1.

Part number: M27/70 - (dash number from table I)

(B) Substitution data (information only). This military transformer is substitutable for FSCM 70674, part number 324-5H. This in no way implies that the manufacturer's part is substitutable for the military item. This information is not intended to be complete

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent
DLA - ES

(Project 5950-0562-16)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-27/70B TRANSFORMER, AUDIO FREQUENCY TF5S2122

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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